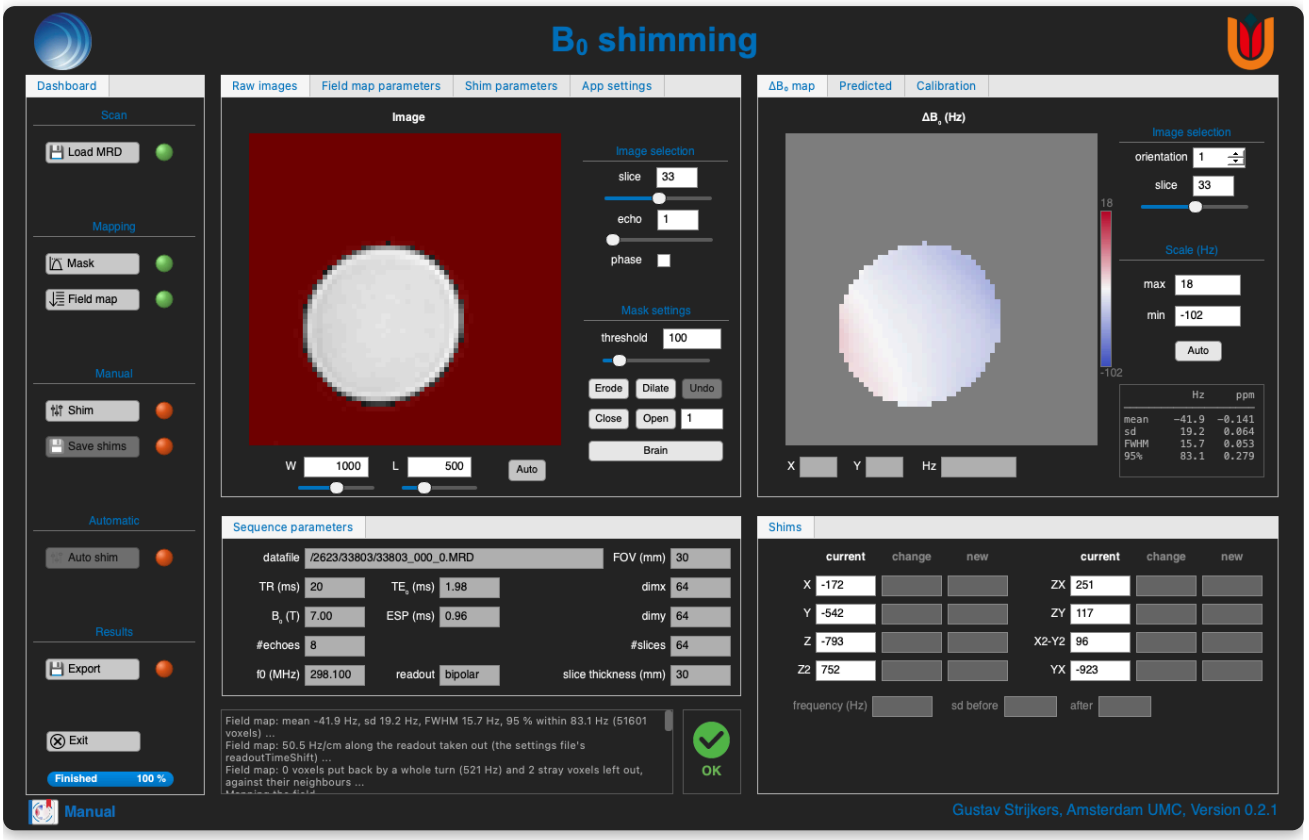



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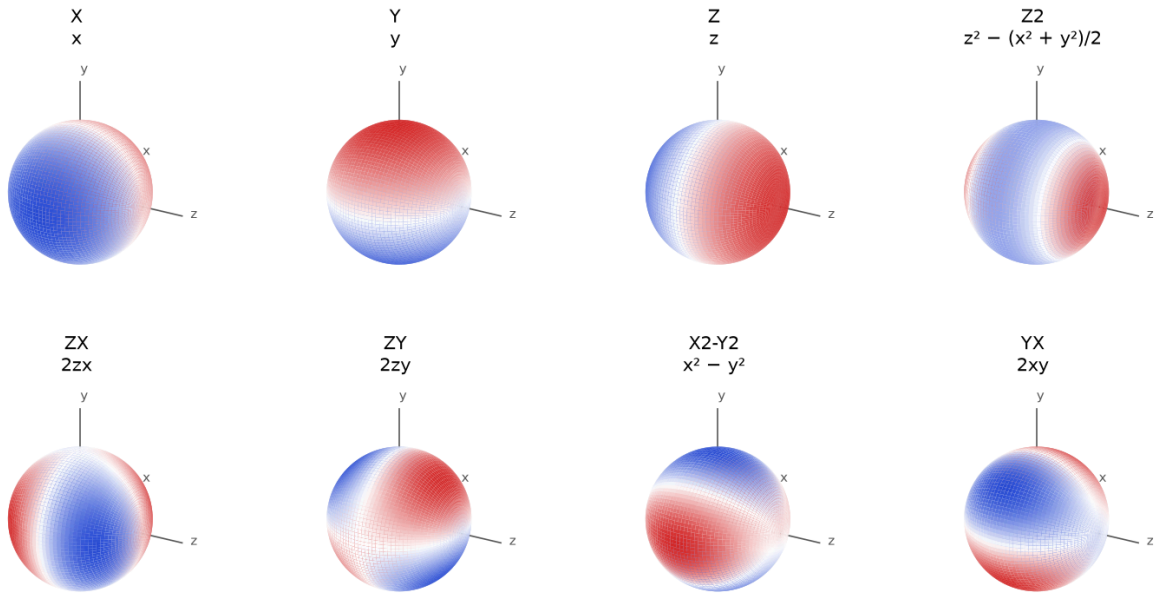
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## 1. Overview

SHIMapp is an application with a graphical user interface for image-based $B_0$ shimming on an MR Solutions preclinical MRI system. It maps the field ($\Delta B_0$, in Hz) of a multi-echo gradient-echo scan, and calculates new first- and second-order shim settings (X, Y, Z, Z2, ZX, ZY, X2-Y2, XY) and a frequency offset that make the field in the region of interest as flat as the shims allow. It shows the field the new shims are predicted to leave, and saves the new shims as a shim file.

The current shims are read from the scan's own folder. What each shim channel does to the field comes from a calibration, made once with a phantom, which can keep learning from every shim round.

**How shimming works.** The magnet's field is never quite the same everywhere in the sample. Each shim channel adds a small field of its own shape to it: X, Y and Z tilt the field along one direction, the five second-order channels bend it (below). SHIMapp maps the field, finds the setting of every channel whose shapes, added up, cancel as much of the measured field as they can, and shows what is predicted to remain: what no combination of the shapes can follow, such as sharp changes near air or bone, stays.




Each channel changes the field in its own shape (a positive setting; red higher, blue lower). z along the bore, y up.

SHIMapp is a stand-alone application: everything it uses is inside it, and nothing else has to be installed.

## 2. Contact

SHIMapp and this manual are written by:

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## 3. Disclaimer

This is a beta version, the first of a new application. However, this does not warrant that the functions contained in the program will meet your requirements or that the operation of the program will be uninterrupted or error-free. Check the shims it proposes before setting them on the scanner. In case of questions or issues, please contact Gustav Strijkers.

## 4. Installation

### macOS

Requirements: a Mac with Apple silicon (M1 or later), macOS 26 or later.

SHIMapp comes as a disk image, `SHIMapp-0.3.4-macOS-arm64.dmg`. Nothing else has to be installed.

1. Open the disk image and drag **SHIMapp** onto the Applications folder in its window.
2. Open it from Applications. The app is not signed with an Apple Developer ID, so the first time macOS refuses it ("Apple could not verify ..."); click **Done**. Then open **System Settings > Privacy & Security**, scroll down to the line saying SHIMapp was blocked, click **Open Anyway** and confirm. From then on it opens as any other app.

   Instead of step 2, the same in Terminal, once:

   ```
   xattr -dr com.apple.quarantine /Applications/SHIMapp.app
   ```

### Windows

Requirements: a PC with Windows 10 or 11, 64-bit.

SHIMapp comes as an installer, `SHIMapp-0.3.4-setup.exe`. Run it; when Windows stops it ("Windows protected your PC"), click **More info** and then **Run anyway**. To remove the app, use **Uninstall SHIMapp** in the Start menu, or **Settings > Apps**.

## 5. Running the software

Start SHIMapp as any other application. While it loads, its logo is shown. The message box says what each step did; the newest message is at the top. The **STATUS** picture shows whether the app is busy, ready, or has a warning or an error. The **Manual** button at the bottom left opens this manual.

## 6. The scan

SHIMapp maps the field from a multi-echo gradient-echo scan (the scanner's FLASH with several echoes), 3D or 2D multi-slice. Field maps for shimming do not need fine detail; they need signal and short echo spacing. A 3D scan, as a start:

| | |
|---|---|
| Sequence | 3D multi-echo gradient echo, fully sampled, 1 average |
| Echoes | 6 to 8, the first as early as possible |
| Echo spacing | as short as the scanner allows; at 7 T about 1.0 ms or 2.0 ms |
| Bandwidth | high (100 kHz or more) |
| TR, flip angle | about 15–20 ms, 8–12° |
| Fat saturation | not needed: at an echo spacing of about 1.0 ms (or 2.0 ms) fat and water are in phase at every echo; in vivo it may be on, at the cost of a longer TR |
| Mouse brain | FOV 20–24 mm, matrix 64 × 64 × 64 |
| Mouse head | FOV 25–30 mm, matrix 64–80 in each direction |
| Rat head | FOV 40 mm, matrix 80 × 80 × 80 |

A 2D multi-slice scan works as well, and **Auto shim** repeats it with the same slices. As a start (tried on the 7 T):

| | |
|---|---|
| Sequence | 2D multi-slice multi-echo gradient echo, fully sampled, 1 average |
| Echoes | 8, the first at about 2 ms, about 1.1 ms apart |
| Bandwidth | 133 kHz (sample period 7.5 µs) |
| TR, flip angle | 500 ms, 30° (about half a minute) |
| Slices | 16 of 1.3 mm, 0.1 mm apart, over the region to shim and a little beyond |
| In plane | matrix 64 × 64, the FOV as for the 3D scan |
| Fat saturation | on in vivo: at an echo spacing of 1.1 ms fat and water are not in phase |

The 3D scan gives the field along the slice direction better; in 2D keep the slices thin and close together.

**Measure several slices or a volume.** A single slice shows the field in its own plane only, so what the shims do through the slice cannot be measured. SHIMapp then leaves those shims as they are (for a transverse slice Z, ZX and ZY), shims that one slice with the others, and the message box says so. To shim a volume, map it with several slices or with the 3D scan.

An accelerated 3D scan (with a sampling table: PE1 order NonRegLUT) works as well: SHIMapp reconstructs it with compressed sensing (**compressed sensing**, section 9). Its sampling table (the LUT file beside the scan) may have any name. On the scanner computer SHIMapp replaces the scanner's images of such a scan by those of its own reconstruction (Step 1).

Plan every shim scan of a study the same way (the same orientation and offsets), and plan the calibration scans the same way too.

**Let the scanner set the frequency (f0) automatically before every shim scan**, 3D and 2D multi-echo gradient echo alike, the calibration scans included. A frequency that changes from scan to scan does not affect SHIMapp's maps or its calibration.

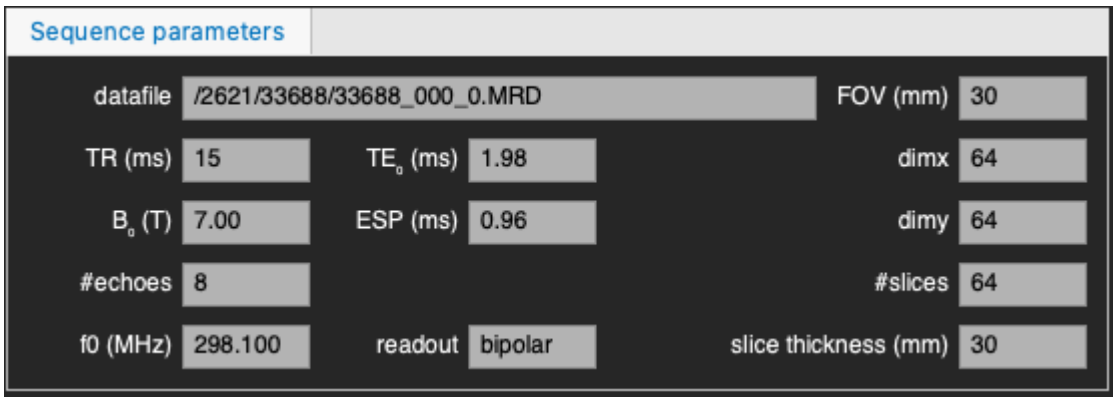
## 7. Basic operation

The **Dashboard** at the left holds the steps, from top to bottom, in sections: **Scan** (**Load MRD**), **Mapping** (**Mask**, **Field map**), **Manual** (**Shim**, **Save shims**), **Automatic** (**Auto shim**) and **Results** (**Export**). The lamp beside each step is green when it is done, yellow while it runs, and red when it is not done yet.




### Step 1: Loading a scan

Press **Load MRD** and choose the scan's MRD file. On the scanner computer (Windows) the first dialog of a session opens in the newest exam of the scanner's MRD folder; after that, in the folder last loaded from. SHIMapp reads the scan and makes the images of its echoes. The **Sequence parameters** below the images show the scan: the data file, TR, the field ($B_0$) and the scanner's frequency (f0), the first echo time ($TE_0$) and the echo spacing (ESP), the number of echoes, the readout (monopolar or bipolar), the field of view, the matrix and the slab or slice thickness. They are shown only.



The shims the scan was made with are read from the scan's folder and shown in the **Shims** panel at the bottom right, under **current**.

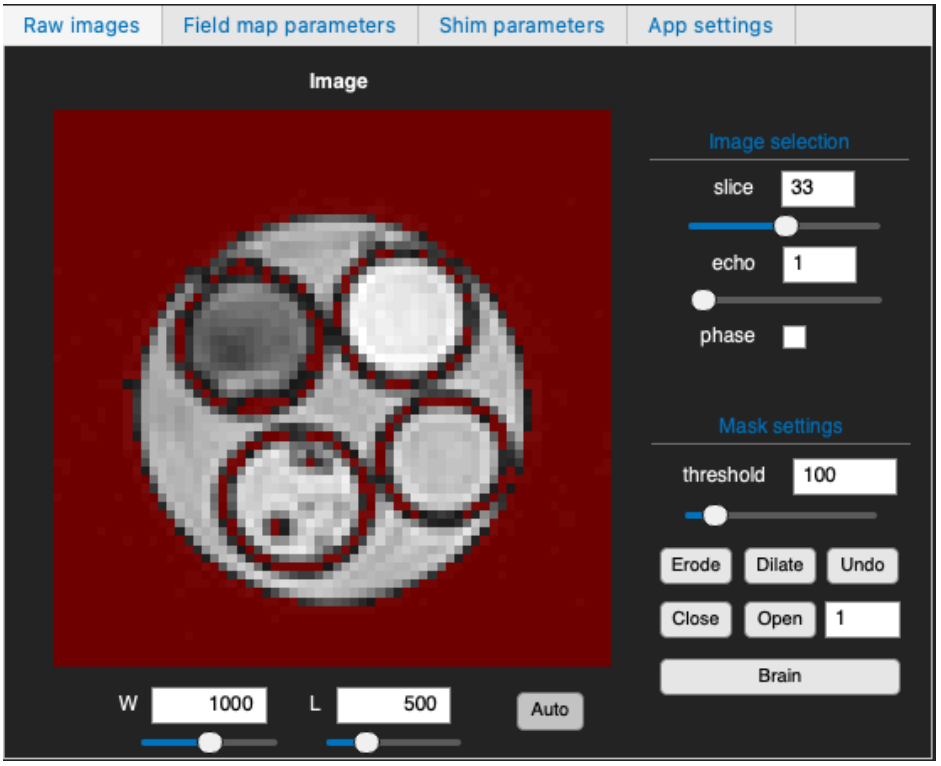
SHIMapp then maps the field (Step 4) and calculates new shims (Step 5) by itself, as if **Field map** and **Shim** were pressed. From there, save the new shims (**Save shims**, Step 6) or let **Auto shim** make the rounds (Step 8). A scan without its shim file beside it is mapped only: enter the current shims and press **Shim**.

**The scanner's images of an accelerated 3D scan.** On the scanner computer SHIMapp replaces the scanner's images (SUR and DICOM files) of a scan with a sampling table by those of its own reconstruction, one per slice and echo, once the scanner has finished writing them and no **Auto shim** run is under way. It does this by itself while you go on, one scan after the other; the message box says when it starts and when it is done. It leaves `shimapp_<scan>.MRD` and `.rpr` beside the scan. A scan is replaced once. The settings file's `replaceScannerImages` (section 9) turns it off.

A change of **k-space lines** (the Field map parameters tab) makes the images, the field map and the new shims again.

### Step 2: The images

The **Raw images** tab shows the images of the echoes, one slice at a time. **slice** and **echo** choose the image; the mouse's scroll wheel over the image steps through the slices as well. **phase** shows the phase the field gives the echo instead of the magnitude, in the mask only: once the field map is made, 2π × the field × the echo time (a well-shimmed object looks even); before, the phase since the first echo read in the same direction (nothing yet on the first echoes). **W** and **L** set the window and level of the magnitude; **Auto** sets them to the image.



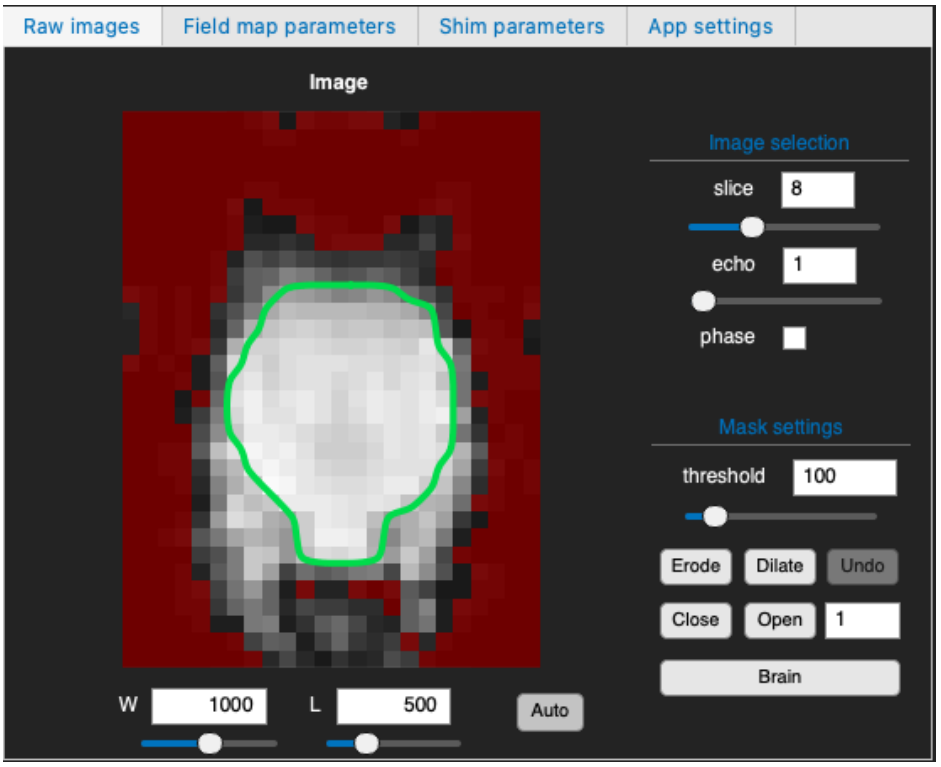
### Step 3: The mask

The voxels outside the mask are shown in red. When a scan is loaded, the mask holds the voxels above a tenth of the image's bright end. **threshold** (or its slider) sets it; **Mask** makes the mask again at that threshold. **Erode**, **Dilate**, **Open** and **Close** change the mask by the number of voxels in the field beside **Open** (in a 2D multi-slice scan, within each slice); **Undo** takes the last change back. After a change of the mask, press **Field map** and **Shim** again.

The field is mapped, and the shims are fitted, in the mask.

**Shimming a part only.** Two ways restrict the shim to a part of the mask; the one chosen last applies, and it is drawn as a thick green line on the raw images and on both maps:

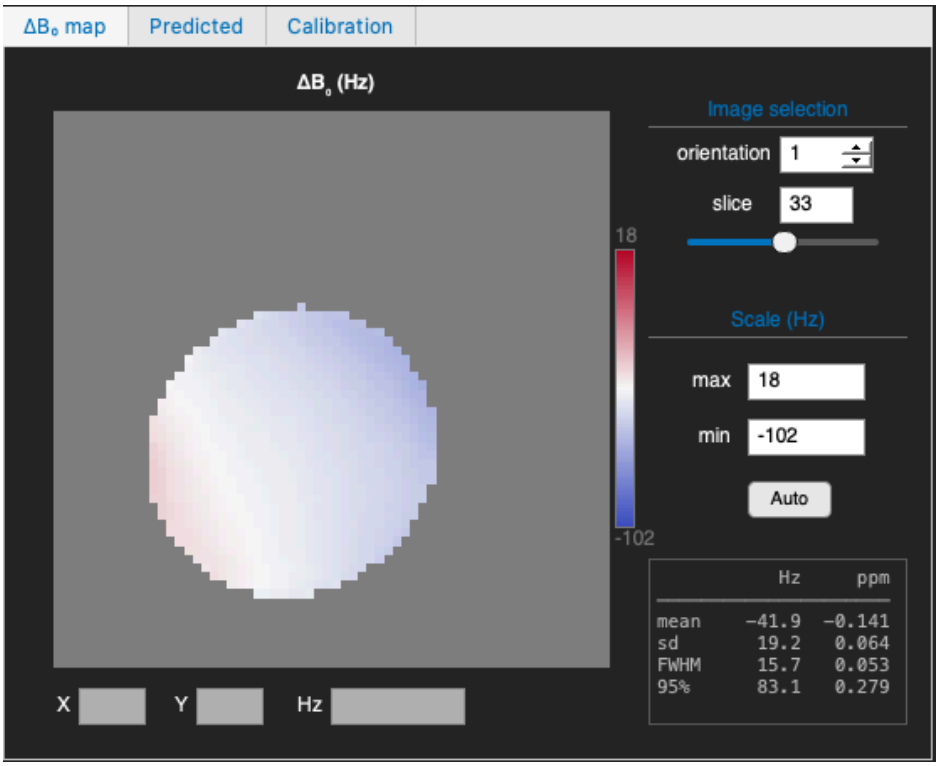
- **Brain** finds the brain: a standard mouse or rat brain is fitted to the scan (mouse or rat by the size of the head), and the shims are fitted to the brain only. It takes a few seconds; the message box gives the brain's volume and how well the standard brain fitted. A poor fit is reported as a warning: check the outline, or draw a box instead.
- **A box:** drag a rectangle over the raw image with the left mouse button. It reaches through as many slices as its shorter side is long, centred on the slice shown; it is drawn solid on those slices and dashed on the others.




A double-click on the raw image removes the box or the brain: the whole mask is shimmed again. Neither changes the mask or the field map, so **Shim** can follow at once. The next scan keeps them when its images are of the same size.

### Step 4: The field map

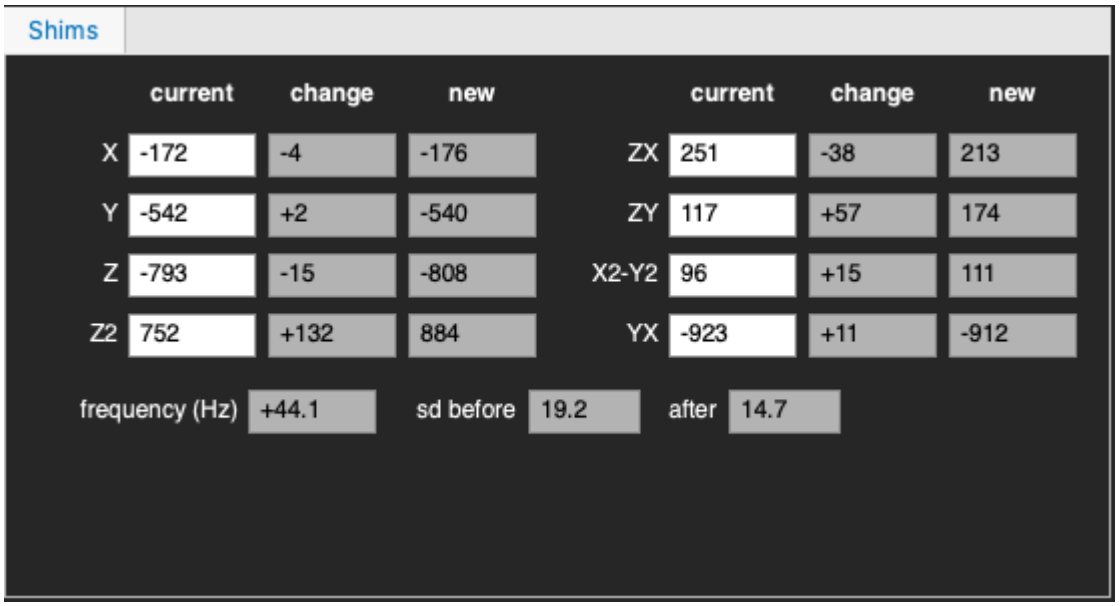
The field is mapped when the scan is loaded; **Field map** maps it again, after a change of the mask or of the Field map parameters. The **$\Delta B_0$ map** tab shows the field of the slice chosen in Hz, from blue (below the scanner's frequency) through white to red (above it), between **min** and **max** of the **Scale**. The scale is centred on the map's mean in the shim region (the frequency, which the frequency offset of **Shim** takes away), equally wide on both sides and the same width for both maps: **Auto** fits it to the field's largest deviations, never narrower than ±50 Hz, so a well-shimmed region looks pale; typing **max** or **min** sets the width for both maps. **orientation** shows both maps in the scan's own slices (1) or in the two planes across them (2 and 3, upright: the magnet's vertical up the screen); **slice** then steps through that plane. A click on the map shows the voxel's position and field under the map, and the scroll wheel over it steps through the slices of both maps. The box at the right gives the field's mean, standard deviation, full width at half maximum and the range of the middle 95 % of the voxels, in Hz and in ppm (of the scanner's frequency; headed **brain** or **box** when one is chosen), in the shim region: the mask, less the voxels whose field could not be measured reliably, within the brain or the box when one is chosen. A voxel whose field is a whole phase turn off its neighbours' is put back, and a voxel that strays far from its neighbours is left out; the message box says how many.




### Step 5: New shims

New shims are calculated once the field is mapped; **Shim** calculates them again, after a change of the brain or the box, the current shims or the Shim parameters. Shims whose effect the region cannot show (in a single slice, those acting through it) are left as they are (section 6). With the standard calibration not yet corrected by a shim round (section 8) it makes only half of the change it calculates, to be safe; the message box says so. The **Shims** panel then shows, for every channel, its **change** and its **new** setting,

with the **frequency (Hz)** offset that centers the field and the field's standard deviation in the shim region **before** and **after**. The **Predicted** tab shows the field the new shims are predicted to leave, on the same scale as the field map.




The **current** settings are the ones the scan was made with: the scan loaded brings its own shims (its `clinical.shm`). The panel shows the eight shim channels; the file's B0 and Z0, which **Shim** does not change, stay as they are. They can be typed in or changed. Which channels are used is set in the **Shim parameters** tab, which also shows the lowest and highest setting each channel may have (section 9).

### Step 6: Saving the shims

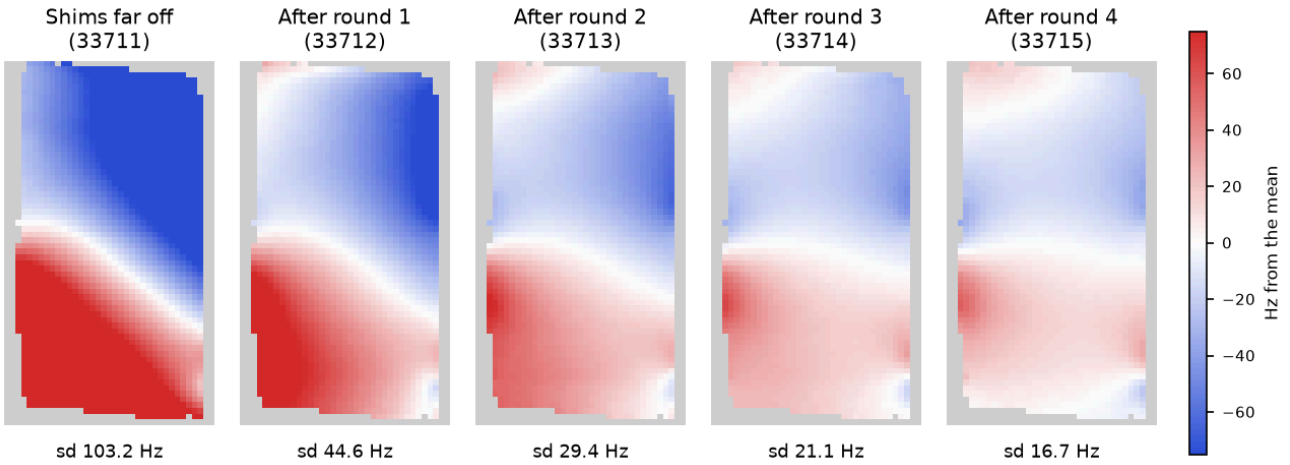
**Save shims** puts the new settings on the scanner. On the scanner computer (Windows) it writes them into the scanner's shim file, `c:\smis\calib\clinical.shm`, with which its next scan is made: the shimmed channels with their new settings, the others (B0, Z0) as the file had them. The file as it was is first kept in the app's folder (`clinical <date> <time>.shm`). Elsewhere, or where that folder is not there, a dialog saves `clinical.shm` where you choose, to copy onto the scanner. Either way the new settings become the current ones. The file is in the scanner's own format (the same channels, numbers and line endings).

### Step 7: The next shim round

After **Save shims**, scan again with the same protocol and load the new scan (or let **Auto shim** make the rounds, Step 8). Its field map checks the round: the message box says whether the scan was made with the shims saved (its own `clinical.shm` says so) and gives the field's spread before, as predicted, and now. With **learn** on (**Options** in the **Shim parameters** tab; off at every start), the calibration also learns from the change between the two scans: what the shim change did to the field; the message box says how well it had been predicted.

Learning happens at **Field map** only, when the scan loaded before this one was mapped too, is planned the same way and was made with other shims; otherwise the message box says why nothing was learned. Mapping a scan again replaces what it taught, so a shim round never counts twice. A calibration that already predicts the change well changes little.

**A shim that acts unexpectedly.** When a shim round changes the field clearly otherwise than the calibration predicts, and as it would if two shims were swapped (for example X and Y) or one, or all of them, worked the other way round, the message box warns and says which, and the calibration in use (`calibration.json`) is changed to match; `default_calibration.json` stays as it was. This is checked at every field map after a shim change, with **learn** on or off, and in the rounds of **Auto shim**, which then go on. If nothing changed on the scanner, check the shims' cabling and the scan. When the change only looks somewhat like that, the message box says so and the calibration is left as it is: two scans with that shim alone changed, a step each way, tell, and **Auto shim** makes such a test by itself (Step 8).




### Step 8: Automatic shim rounds

On the scanner computer, **Auto shim** (the **Automatic** section) takes the shim rounds over. Load the shim scan made with Preclinical Scan (its field of view, and in 2D its slices, planned as usual): its field is mapped and new shims calculated. Choose the brain or a box if you want (and press **Shim** again); then press **Auto shim**.

Round after round, SHIMapp sets the new shims, sets the frequency, makes the same scan again, maps its field with the same mask (and the same brain or box) and calculates new shims. The message box gives each round's field spread before, as predicted and as measured, and the images and the **$\Delta B_0$ map** show the newest round: the map in front, in the slice shown and on the same scale throughout, so that the rounds compare. The rounds end when the next shims are predicted to improve the field by less than a set amount, after a set number of rounds, or when a round leaves the field worse than the best round so far. SHIMapp then sets the best round's shims, saves them in the scanner's shim file (`clinical.shm`; the file as it was kept as **Save shims** keeps it) and sets the frequency once more. The scanner's next scans are made with them. The message box then says **AUTO SHIM FINISHED**, with the field's spread at the start and at the end, the rounds made and why they ended, and where the shims are saved.

When a round looks somewhat, but not clearly, as if a shim were swapped with another or worked the other way round (Step 7), the next two rounds are a test: that shim alone is set a step above and then a step below its setting, each step large enough to show it clearly; near one of its limits, one and two steps away from that limit instead (a test never sets a shim beyond its limits). The calibration is changed

only when both steps show the same; when neither does, the shim acts as calibrated; otherwise the calibration is left as it is. The rounds then go on from the second test scan. A run makes at most two such tests, and the test scans do not count in the most rounds.

While the rounds go on, the button reads **Stop**: pressing it stops the scan under way, and the rounds end in the same way, with the best round's shims. With **learn** on, each round teaches the calibration as well (section 8).

The rounds' scans go to the scanner's temporary folder (`c:\smis\dev\Temp`), each run in a folder of its own, and are removed when the rounds end; they are not added to the exam. The most rounds, the smallest improvement worth another round and the frequency step before each round are set in the settings file (section 9). **Auto shim** works on the scanner computer only, with the scanner software running.

### Step 9: Exporting

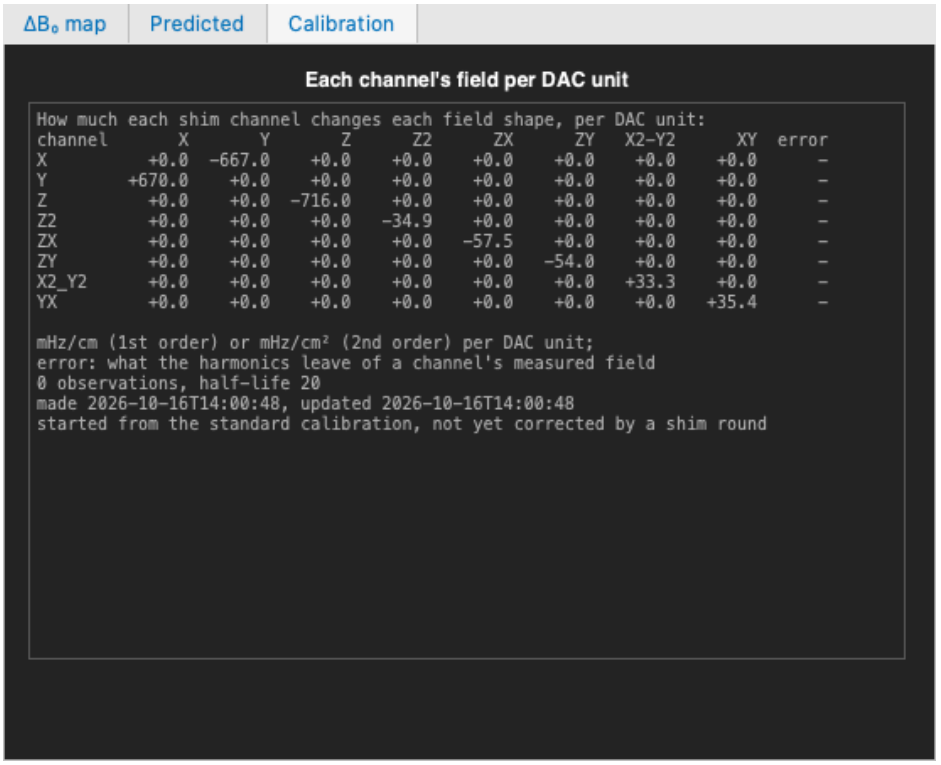
**Export** saves the field map, the predicted field and the shim region (NIfTI), with a report of the shims and the field's statistics (`report.json`), in a new folder where you choose.

**Exit** closes the app.

## 8. The calibration

The calibration tells SHIMapp how much each shim channel changes the field per unit of its setting. **Shim** needs it to turn the measured field map into new settings. It is shown in the **Calibration** tab: for each channel, its field in each of the first- and second-order shapes, and how much of the channel's measured field these shapes leave unexplained (**error**). The **calibration** field of the **Shim parameters** tab says which calibration is in use: the standard one (and how many shim rounds have corrected it) or the file's name.

**The two calibration files.** The app's folder (section 9) holds the calibration in use, `calibration.json`, and its backup, `default_calibration.json` (the standard calibration as it comes). SHIMapp starts from `calibration.json`, and every shim round it learns from (**learn** on) updates it. To start over, delete `calibration.json`: the next start makes it again from `default_calibration.json`. To use a calibration of your own, edit `calibration.json`, or put your file in its place, while the app is closed.



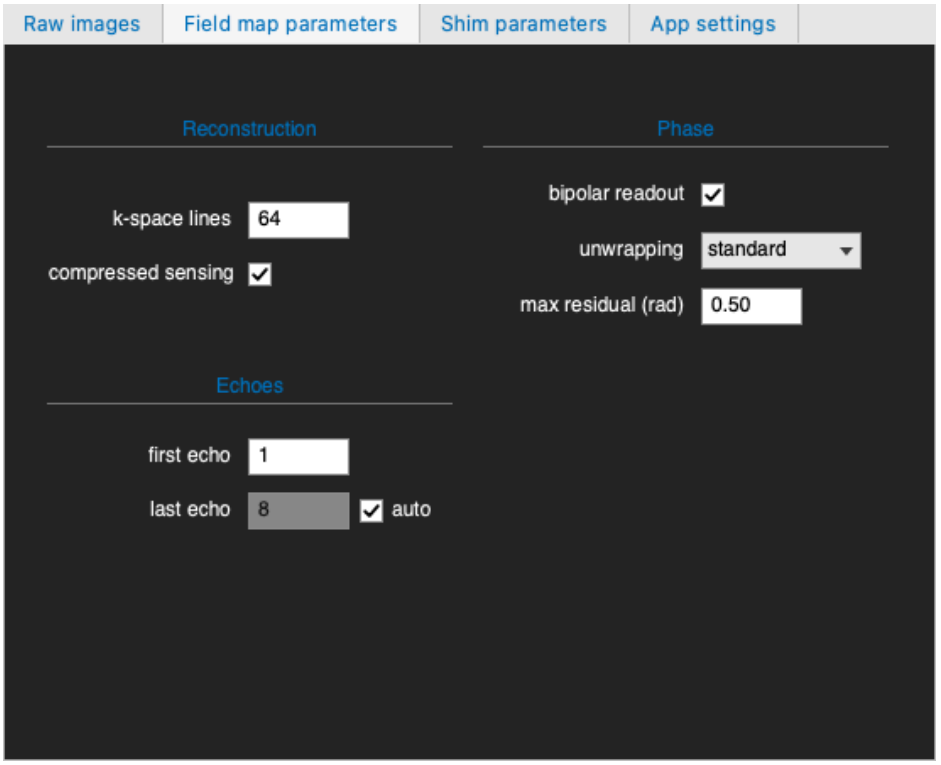

**Making a calibration.** Use a large uniform phantom that fills the coil, shimmed first. Scan it with the shim scan's protocol: once as a reference, then once for each channel (X, Y, Z, Z2, ZX, ZY, X2-Y2, XY) with that channel changed by an amount that changes the field over the phantom by about 100 to 300 Hz. The command line makes the calibration from these scans, reading each scan's shims from its folder: `SHIMapp --calibrate REFERENCE.MRD SCAN.MRD ... --out my-calibration.json` (on macOS `SHIMapp.app/Contents/MacOS/SHIMapp`). Put it in place of `calibration.json` while the app is closed.

A calibration keeps the planning of the scans it was made from, and follows each scan's own planning (its angles, offsets, 3D or 2D, phase orientation) when it shims or learns: a scan may be turned, and its field of view shifted off the isocenter.

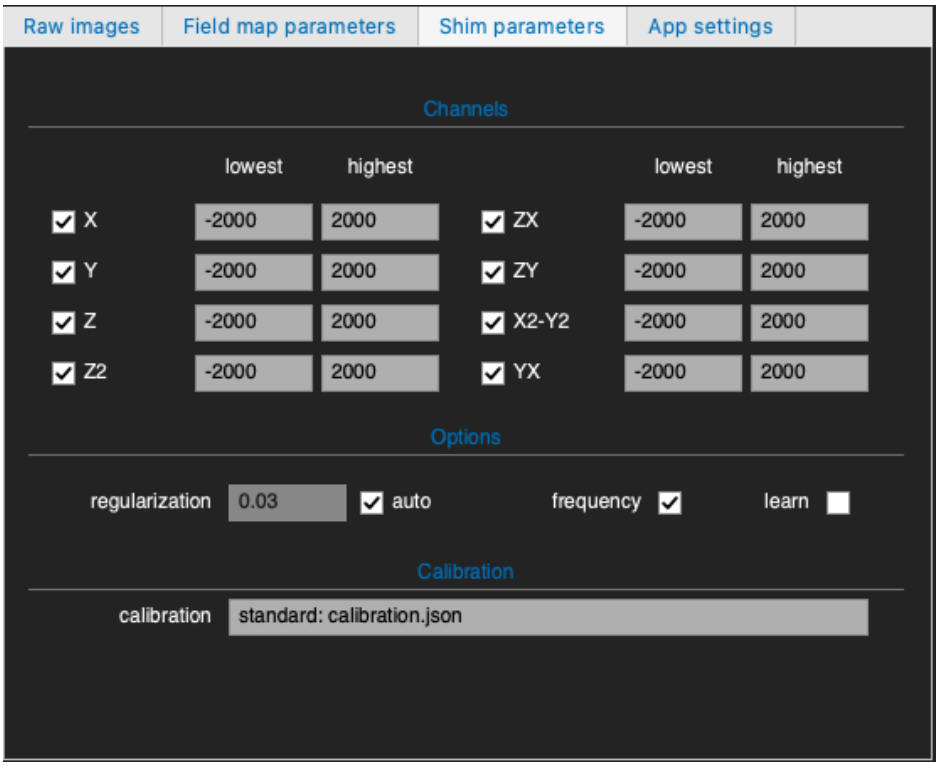
**The standard calibration** is the first default: the strengths of the 7 T's shim channels, measured on a water phantom with each channel changed on its own, each channel on the field shape it makes (0.67 to 0.72 Hz/cm per unit for X, Y and Z, and 0.033 to 0.058 Hz/cm² for the second order). Their small crosstalk is left out, and the first round makes half the calculated change. With **learn** on, every round corrects the calibration with what the shims really did. The standard calibration follows each scan's own planning (its view, angles, offsets and phase orientation), so it serves turned and shifted scans as well. Only for a scan whose slices are planned in different orientations, and for an EPI scan, **Shim** asks for a calibration made in that kind of planning, in place of `calibration.json`.

A new version of SHIMapp starts from the calibration the version before used.

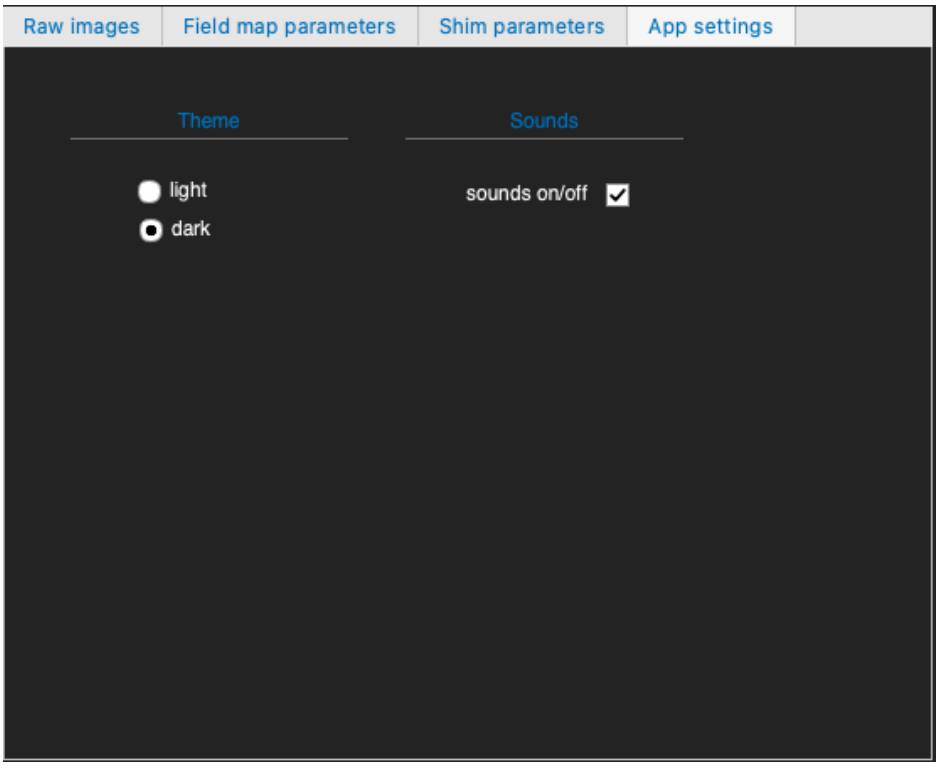
## 9. The settings




**Field map parameters: k-space lines** (the matrix of the field map's images, at most; a change makes the images again); **compressed sensing** (on as it comes: an accelerated 3D scan, one with a sampling table, is reconstructed with compressed sensing up to the **k-space lines**, its voxels no smaller than 0.47 mm (the settings file's `compressedSensingVoxel`); off, only the part of k-space it sampled fully is used, which gives coarser images; a fully sampled scan is not changed by it; a change makes the images again. Compressed sensing is much faster with the BART toolbox installed (on Windows, in WSL); the progress bar shows how far it is); **bipolar readout** (set from the scan); **unwrapping** (standard or quality); **max residual (rad)** (how far a voxel's phase may stray before it is left out of the shim region); **first echo** and **last echo** (the echoes used; with **auto**, on as it comes, the last echo is chosen at every **Field map**: the echoes until the signal has faded, at least 8, as the message box says).




**Shim parameters: Channels** (the channels the shims may change), each with its **lowest** and **highest** setting beside it (from the settings file, below: shown only, they cannot be changed here); **Options**: **regularization** (keeps the changes small when the region is too small to tell channels apart; 0 for none; with **auto**, on as it comes, SHIMapp chooses it at every **Shim**: the largest that costs the shim practically nothing, shown in the field), **frequency** (a frequency offset as well) and **learn** (learn from every shim round, section 8; off at every start); **Calibration**: the **calibration** in use (section 8).




**App settings:** the **Theme** of the window (light or dark), and the **sounds** of warnings and errors.

**The app's folder** (macOS: `~/Library/Application Support/SHIMapp/0.3.4p`; Windows: `%APPDATA%\SHIMapp\0.3.4p`) holds these settings (`shimappSettings.json`), a log of every session (`shimapp-log-<date-time>.txt`, every message with its time; the latest 100 are kept), the two calibration files (`calibration.json`, in use, and `default_calibration.json`, section 8), the default shims (`default.shm`: the current shims shown before a scan is loaded), the scanner's shim files kept by **Save shims**, and the shim files you save there. The shim and calibration dialogs open in it. The settings file also names the scanner's shim file, `scannerShimFile` (`c:\smis\calib\clinical.shm` as it comes).

**Each channel's limits** are set in the settings file, as `shimLimits`: for each shim channel its lowest and highest setting in DAC units, `[-2000, 2000]` for every channel as they come. **Shim** never sets a channel beyond its own limits (a channel already beyond them may only come back). Change them in the file while the app is closed; a channel left out, or written wrongly, keeps `[-2000, 2000]`.

**Which channel makes which shape** is set in the settings file, as `channelSwap`, for a scanner on which a shim channel makes another channel's shape: each such channel with the channel whose shape it makes, `{"ZX": "ZY", "ZY": "ZX"}` as it comes, `{}` when every channel makes its own shape. A pair named once (`{"ZX": "ZY"}`) is swapped both ways. The channels keep the scanner's names and settings in SHIMapp, so the shims it shows are the ones to set on the scanner. Change it in the file while the app is closed; an entry written wrongly is not used, and the message box says so.

**How fast what the calibration learned fades** is set in the settings file, as `halfLife`: the number of shim rounds after which what a round taught weighs half, `20` as it comes; `0` weighs every round alike. Change it in the file while the app is closed; a value written wrongly keeps `20`, and the message box says so.

**The automatic shim rounds** (Step 8) are set in the settings file, as `autoShim`: `maxRounds`, the most rounds after the first scan (`5` as it comes); `minGainHz`, the smallest improvement of the field's spread (sd, in Hz) the next round must promise (`1`); and `frequencyStep`, whether the frequency is set before each round (`true`; it is set at the end in any case). Change them in the file while the app is closed; a value written wrongly keeps the standard one, and the message box says so.

**How fine compressed sensing reconstructs** is set in the settings file, as `compressedSensingVoxel`: the smallest voxel, in mm along each phase direction, an accelerated 3D scan is reconstructed to, `0.47` as it comes (the voxels of the 3D shim scan: 30 mm in 64 lines). A field of view of 18 x 16 mm then gets 38 x 34 lines. Smaller voxels give finer but noisier field maps and take longer. Change it in the file while the app is closed; a value written wrongly (outside 0.05 to 5) keeps `0.47`, and the message box says so.

**The scanner's images of an accelerated 3D scan** (Step 1) are replaced as it comes; the settings file's `replaceScannerImages`, `false`, leaves them as the scanner made them. Change it in the file while the app is closed; a value other than `true` or `false` keeps `true`, and the message box says so.

**The readout's time shift** is set in the settings file, as `readoutTimeShift`: one value for 3D scans and one for 2D scans, in µs, `{"3D": 1.09, "2D": 1.16}` as it comes (the 7 T's). From it SHIMapp works out, for each scan's field of view, bandwidth and echo spacing, the gradient along the readout that it takes out of the field map (the message box gives it in Hz/cm), so that one set of shims gives the same field map in every planning, whichever way the readout runs; `0` leaves a map as measured. Change it in the file while the app is closed; a value written wrongly keeps the standard one, and the message box says so.

## 10. Credits

SHIMapp uses Python, NumPy, SciPy, scikit-image, numba, PySide6 (Qt), pyqtgraph, nibabel, SigPy and PyWavelets. Its window is built as those of the other MRI apps of Amsterdam UMC (Retrospective, P2ROUD, T1mapp, T2mapp, ADCmapp, DSCmapp).

The standard brains of the **Brain** button are made, changed, from:

- **Mouse:** the in-vivo C57BL/6J mouse brain atlas of Brookhaven National Laboratory and the National High Magnetic Field Laboratory (Ma et al., NeuroImage 2005 and 2008; supported by NIH grants R01 EB 00233-04 and P41 RR16105 and the National High Magnetic Field Laboratory), https://www.nitrc.org/projects/c57bl_mr_atlas/, licensed CC BY-NC-SA 4.0; the mouse template in SHIMapp is shared under the same licence.
- **Rat:** the SIGMA rat brain templates and atlases, version 2.0 (Barrière et al., Nature Communications 2019), https://zenodo.org/records/10635831, licensed CC BY 4.0; SHIMapp's rat template is changed from it (resampled and cropped).